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Line-of-sight detection apparatus and image capturing apparatus

Abstract

A line-of-sight detection apparatus includes an acquisition unit configured to acquire an eyeball image, an irradiation unit configured to irradiate an eyeball with light in an infrared wavelength band, forming a plurality of point-shaped optical images on the eyeball image, and a calculation unit configured to calculate line-of-sight information based on the eyeball image and the plurality of point-shaped optical images, wherein the irradiation unit includes at least one single light source configured to emit the light in the infrared wavelength band, and a dividing element configured to divide the light emitted from the single light source into a plurality of beams of the emitted light to form the plurality of point-shaped optical images.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11237628	12/2021	Sharma	N/A	G02B 27/1086
2011/0019874	12/2010	Jaervenpaeae	N/A	N/A
2012/0133754	12/2011	Lee	348/78	H04N 21/4223
2019/0253700	12/2018	Tornéus	N/A	G02B 27/0093
2020/0012105	12/2019	Liu	N/A	H04N 23/20
2020/0092534	12/2019	Eckhouse	N/A	G06T 7/0012
2020/0305712	12/2019	Kojima	N/A	A61B 3/113

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H02004313	12/1989	JP	N/A
2002119478	12/2001	JP	N/A
2010017407	12/2009	JP	N/A
2019-139743	12/2018	JP	N/A
2020036741	12/2019	JP	N/A
2020204914	12/2019	JP	N/A
2023021724	12/2022	WO	N/A

OTHER PUBLICATIONS

The above Foreign Patent Documents were cited in a Japanese Office Action that issued on Jun. 6, 2025, which is enclosed, with an English translation, that issued in the corresponding Japanese Patent Application No. 2021-157238. cited by applicant

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Background/Summary

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The present invention relates to a line-of-sight detection apparatus that can detect a line of sight.
Description of the Related Art
(2) Line-of-sight detection techniques have been studied and practical advancements have been

made in fields such as detection of the line-of-sight of a driver of an automobile or the like, study on human line-of-sight behavior, and techniques for supporting handicapped persons. A corneal reflection method is one of a plurality of methods proposed for line-of-sight detection, and is advantageous in terms of accuracy and downsizing. The corneal reflection method is a method of calculating line-of-sight information from a near infrared eyeball image, and is based on the arrangement of a Purkinje image in the eyeball image.
(3) To acquire the line-of-sight information using the corneal reflection method, a plurality of Purkinje images need to be formed in the eyeball image. For example, in Japanese Patent Laid-Open No. 2019-139743, this is realized by arranging a plurality of infrared light sources on eyeglasses.
(4) Still, while there is demand for an apparatus that is smaller and more accuracy for line of sight detection for the line-of-sight detection technique, it is difficult to achieve such smaller and more accurate apparatus with known techniques, since a large number of light sources are required and a degree of freedom in the arrangement of the light sources is low.
SUMMARY OF THE INVENTION
(5) The present invention has been made in view of the problem described above, and provides a line-of-sight detection apparatus that has a small size and can detect the line of sight with high accuracy.
(6) According to a first aspect of the present invention, there is provided a line-of-sight detection apparatus comprising: at least one processor or circuit configured to function as: an acquisition unit configured to acquire an eyeball image; an irradiation unit configured to irradiate an eyeball with light in an infrared wavelength band, forming a plurality of point-shaped optical images on the eyeball image; and a calculation unit configured to calculate line-of-sight information based on the eyeball image and the plurality of point-shaped optical images, wherein the irradiation unit includes at least one single light source configured to emit the light in the infrared wavelength band, and a dividing element configured to divide the light emitted from the single light source into a plurality of beams of the emitted light to form the plurality of point-shaped optical images.
(7) According to a second aspect of the present invention, there is provided an image capturing apparatus comprising: an image capturing device configured to capture an image of a subject; a display apparatus configured to be able to display the image captured by the image capturing device; and the line-of-sight detection apparatus described above.
(8) According to a third aspect of the present invention, there is provided a line-of-sight detection apparatus comprising: at least one processor or circuit configured to function as: an acquisition unit configured to acquire an eyeball image; an irradiation unit configured to irradiate an eyeball with light in an infrared wavelength band, forming a plurality of point-shaped optical images on the eyeball image; and a calculation unit configured to calculate line-of-sight information based on the eyeball image and the plurality of point-shaped optical images, wherein the irradiation unit has a micro light source array including micro light sources being arranged in a two-dimensional array.
(9) According to a fourth aspect of the present invention, there is provided an image capturing apparatus comprising: an image capturing device configured to capture an image of a subject; a display apparatus configured to be able to display the image captured by the image capturing device; and the line-of-sight detection apparatus described above.
(10) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a schematic diagram illustrating a configuration of a line-of-sight detection

apparatus according to a first embodiment of the present invention.
(2) FIG. **1**B is a schematic diagram illustrating a configuration of a line-of-sight detection apparatus according to a modification example of the first embodiment.
(3) FIG. **2** is a schematic diagram illustrating an example of an eyeball image.
(4) FIG. **3** is a diagram illustrating an example of a known line-of-sight detection apparatus.
(5) FIG. **4** is a schematic diagram illustrating an example of an eyeball image according to the first embodiment.
(6) FIG. **5**A is a schematic diagram illustrating an eyeball image for describing individual differences in a human eye.
(7) FIG. **5**B is a schematic diagram illustrating an eyeball image for describing individual differences in a human eye.
(8) FIG. **6** is a schematic diagram illustrating Purkinje images with which individual differences in a human eye are difficult to reduce.
(9) FIG. **7**A is a schematic diagram illustrating Purkinje images in a two-dimensional array form on an eyeball image.
(10) FIG. **7**B is a schematic diagram illustrating Purkinje images in a two-dimensional array form on an eyeball image.
(11) FIG. **8** is a schematic diagram illustrating a configuration of a line-of-sight detection apparatus according to a third embodiment.
(12) FIG. **9** is a schematic diagram illustrating a configuration of a line-of-sight detection apparatus according to a fourth embodiment.
(13) FIG. **10** is a schematic diagram illustrating a configuration of an image capturing apparatus according to a fifth embodiment.

DESCRIPTION OF THE EMBODIMENTS

(14) Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention. Multiple features are described in the embodiments, but limitation is not made to an invention that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

First Embodiment

(15) A first embodiment of the present invention will be described below.
(16) Line-of-Sight Detection Apparatus
(17) FIG. **1** is a diagram illustrating a configuration of a first embodiment of a line-of-sight detection apparatus of the present invention. In the present embodiment, a configuration is employed in which the line-of-sight position of an eyeball **102***a* is identified by using a corneal reflection method.
(18) The eyeball **102***a* includes, in the white portion of the eye, an iris portion **102**b including the pupil. A line-of-sight detection apparatus **101** of the present embodiment includes a single light source **101***e* configured to irradiate an eyeball surface (a surface portion including the cornea) with light **101**f in a near infrared wavelength band. The light **101***f* emitted from the single light source **101***e* is divided into light in two directions by an irradiation light dividing element **101***g*, and emitted onto the surface of the eyeball **102***a*.
(19) The irradiation light dividing element **101***g* is not limited to a particular element, and any element that divides light may be used such as a diffractive optical element (DOe) using optical interference, or a micro scanner element.
(20) In the configuration illustrated in FIG. **1**A, the light emitted to the eyeball surface is partially reflected by the eyeball surface, and then is further reflected by a reflection mirror **101***c*, to reach an image sensor **101***a*, serving as an eyeball image acquisition unit, through an imaging lens **101***b*. A point irradiation unit of the present embodiment includes the single light source **101***e* and the

irradiation light dividing element **101***g*.
(21) Line-of-sight information includes a direction/angle of a viewing axis, a position, on a screen region facing the eyeball, where the line-of-sight crosses the screen, and the like. The line-of-sight information is acquired from a line-of-sight information calculation unit **103**, based on the eyeball image acquired by the image sensor **101***a*.
(22) In the present embodiment, the reflection mirror **101***c* is a dichroic mirror that reflects near infrared light and transmits visible light, so that the situation in which a subject can see the scenery through the mirror is realized. Still, the present embodiment is not necessarily limited to a configuration involving wavelength selection using the dichroic mirror. The configuration illustrated in FIG. **1**A should not be construed in a limiting sense, and a configuration illustrated in FIG. **1**B may be employed for example, in which near infrared reflected light from the eyeball is made incident on the image sensor **101***a* through a bandpass filter **101***h* through which only the infrared light passes.
(23) With this configuration, the line-of-sight information calculation unit **103** can obtain the near infrared eyeball image.
(24) An eyeball image **301** illustrated in FIG. **2** is an example of the eyeball image obtained with the configuration illustrated in FIG. **1**A or FIG. **1**B. In the eyeball image **301**, an eyelid portion **305**, a white eye portion **304** of the eyeball therein, and an iris portion **303** therein, and a pupil **302**, which is an opening portion, near the center of the iris portion **303** are captured. The present example is a typical example of an eyeball image obtained by the corneal reflection method.
(25) The light divided in two by the irradiation light dividing element **101***g* forms Purkinje images **306** that are two point-shaped optical images, on the eyeball image **301**. Here, a pair of Purkinje images **306** are formed in a direction substantially parallel to an x direction in FIG. **2**, and are adjusted to sandwich the center of the eyeball image or the pupil region. This allows the line-of-sight information in the x direction to be calculated using the positions of the Purkinje images **306** on the eyeball image, as well as the contour shape and the position of the pupil **302**.
(26) In the present embodiment, by providing the irradiation light dividing element **101***g*, unit downsizing can be achieved. With the known configuration illustrated in FIG. **3**, the unit downsizing is difficult to achieve because a light source corresponding to each Purkinje image is required for forming the plurality of Purkinje images, due to the absence of the irradiation light dividing element.
(27) Through the irradiation of light to form, on the eyeball image, pairs of Purkinje images two-dimensionally arranged at least respectively in the horizontal direction and the vertical direction on the pupil **302** as the center as illustrated in FIG. **4**, the line-of-sight information in at least the x direction and a y direction can be detected. Thus, the two-dimensional line-of-sight information can be detected.
(28) In FIG. **4**, the pair of Purkinje images **306** are used for the line-of-sight information in the x direction, and a pair of Purkinje images **401** are used for the line-of-sight information in the y direction. Note that the pair of Purkinje images **401** are also formed to be opposite to each other with the pupil **302** at the center of the eyeball image interposed in between.
(29) It has been known that the detection accuracy of the line-of-sight detection is largely affected by individual differences in eyes between subjects, that is, persons. In particular, covering amount of the eyeball by the eyelid differs among people, and thus the covering amount largely impacts the accuracy of the line-of-sight detection. FIG. **5**A illustrates an example of an eyeball image of a person whose eyelids are ideally opened. FIG. **5**B illustrates an example of an eyeball image of a person whose eyeball is relatively largely covered by his or her eyelid.
(30) It is relatively common for an eyeball of a person to be covered with the upper eyelid, but it is rare for the eyeball to be largely covered with the lower eyelid. In FIG. **5**A, the captured image clearly includes four of the Purkinje images as intended. On the other hand, a Purkinje image **501** (one of the Purkinje images) that is supposed to be on the +y side is hidden by the upper eyelid in

FIG. **5**B.

(31) Still, even in the case of FIG. **5**B, according to the arrangement of four of the Purkinje images of the present embodiment, the Purkinje image on the −y side is present and only the Purkinje image on the +y side is hidden. Thus, the captured image includes the pair of Purkinje images **306** and one Purkinje image **401**, and three of the Purkinje images are two-dimensionally arranged, whereby the line-of-sight information in the x direction and the y direction can be acquired.

(32) FIG. **6** illustrates a case where Purkinje images paired in the x direction and the y direction are formed, but in each direction, the Purkinje images are not formed opposite to each other with the pupil **302** at the center of the eyeball image **301** interposed in between. The two-dimensional line-of-sight information cannot be obtained in the case illustrated in FIG. **6**, because of four target Purkinje images **601** and **601′**, the Purkinje images **601′** on the +y side are hidden.

(33) As described above, with the line-of-sight detection apparatus of the present embodiment, two pairs of Purkinje images in the x direction and the y direction are each formed to be opposite to each other with the pupil **302** interposed in between, so that the line-of-sight detection can be performed with high accuracy even when vignetting is occurring on the eyeball image due to an eyelid.

(34) With the line-of-sight detection apparatus of the present embodiment, Purkinje images are formed in a two-dimensional array form on the eyeball image as illustrated in FIG. **7**A and FIG. **7**B, so that the line-of-sight information can be acquired with higher accuracy. Examples of the arrangement in the two-dimensional array form include, but are not limited to, a lattice form illustrated in FIG. **7**A, a concentric form as illustrated in FIG. **7**B where Purkinje images are each disposed on a corresponding one of straight lines passing through the center, and the like.

(35) When a plurality of groups of Purkinje images are formed using a plurality of respective single light sources, the impact of stray light and scattered light can be suppressed by varying the time of formation among the Purkinje image groups.

Second Embodiment

(36) Line-of-Sight Detection Apparatus

(37) A line-of-sight detection apparatus of a second embodiment is configured with the point irradiation unit in the line-of-sight detection apparatus **101** of the first embodiment replaced with a two-dimensional micro light source array. A micro light source forming the two-dimensional micro light source array may be a light emitting diode, or a Vertical cavity Surface emitting Laser Diode (VcSeL). Light sources two-dimensionally arranged in-plane parallel to a substrate can be used as the two-dimensional micro light source array. Furthermore, a laser bar stack that is a stack of one-dimensionally arranged edge emitting type laser elements on the substrate can be use as the two-dimensional micro light source array. With such a two-dimensional micro light source array used as the point irradiation unit, an array of point-shaped optical images can be formed on an eyeball image efficiently with a compact configuration.

Third Embodiment

(38) Electronic Apparatus

(39) FIG. **8** is a schematic diagram illustrating a configuration of a line-of-sight detection apparatus **701** according to a third embodiment. The third embodiment relates to a configuration obtained by providing the line-of-sight detection apparatus of the first embodiment with a display apparatus that can display an image.

(40) In FIG. **8**, the line-of-sight detection apparatus **701** includes a display **801** serving as the display apparatus, and a controller **802** that controls what is displayed on the display. In a state where a video or the like is being displayed on the display **801**, an observer can observe the video on the display **801** through the dichroic mirror **101***c*.

(41) In the present embodiment, the line-of-sight information calculation unit **103** calculates the line-of-sight direction of the observer from the eyeball image acquired by the image sensor **101***a*. With the information on this line-of-sight direction transmitted to the display controller **802**, the

line-of-sight position can be displayed on the display in a superimposing manner in real time. With the configuration illustrated in FIG. **8**, a television monitor, a monitor for a personal computer, or the like may be used as the display **801**.

Fourth Embodiment

(42) Electronic Apparatus

(43) FIG. **9** is a schematic diagram illustrating a configuration of a line-of-sight detection apparatus **901** according to a fourth embodiment. A configuration of the fourth embodiment is obtained with the line-of-sight detection apparatus **701** of the third embodiment provided with an eyepiece lens **902**, with which the video on the display apparatus **801** is optimized and delivered to the eyeball.

(44) When the line-of-sight detection apparatus is incorporated in an apparatus formed as a small unit such as a camera electronic viewfinder, a display apparatus for the line-of-sight detection apparatus is small. Thus, in the present embodiment, the eyepiece lens **902** is disposed in front of the display apparatus **801**, so that the observer can comfortably view the video on the small display apparatus **801**.

(45) In the configuration of the present embodiment, two single light sources **101***e* and two irradiation light dividing elements **101***g* are provided, to form the Purkinje images **306** and **401** on the eyeball image using emitted light from the respective single light sources as illustrated in FIG. **4**.

Fifth Embodiment

(46) Image Capturing Apparatus

(47) FIG. **10** is a diagram illustrating a configuration of an image capturing apparatus **1000** including an electronic viewfinder **1020**, according to a fifth embodiment. The line-of-sight detection apparatus **901** is incorporated in the electronic viewfinder **1020**.

(48) In FIG. **10**, the image capturing apparatus **1000** includes an image sensor **1001** configured to capture an image, an MPU **1011** configured to perform processing on various signals, data, and the like for the image capturing apparatus, a liquid crystal display **1013** in charge of input/output to and from the external, an operation switch group **1014**, and a memory **1015**. An image sensor driving circuit **1008** configured to drive the image sensor **1001**, and an image processing circuit **1009** configured to perform processing of an image signal from the image sensor **1001** are further provided.

(49) When a focus detection operation is instructed through a focus detection operation unit **1012** including a button and the like used for an operation for the focus detection, the focus detection is performed based on a captured image signal from the image sensor **1001**. The image sensor **1001** is an element configured to acquire an image, which is different from an image sensor configured to capture the eyeball image provided in the line-of-sight detection apparatus **901** in the electronic viewfinder **1020**.

(50) In the present embodiment, focus detection and focus adjustment are performed using on-imaging surface phase difference method. The focus adjustment is performed by performing focus detection, calculating the driving amount for a focus lens in an imaging optical system **1003** required for focusing, and moving the focus lens to a focusing position.

(51) Image data acquired by the image sensor **1001** through the imaging optical system **1003** is processed by the image processing circuit **1009** as to-be-stored image data, and then is stored in the memory **1015**, such as an SD card, detachably attached to the image capturing apparatus **1000**. Live view data for a state where a shooting button is not pressed is processed to be displayed on the electronic viewfinder **1020** or the liquid crystal display **1013**, and then is displayed on the display. A user viewing the electronic viewfinder **1020** can observe his or her line-of-sight position pointer displayed in a superimposing manner with the video.

(52) With the user's line-of-sight position pointer displayed in real time on the display of the electronic viewfinder **1020**, a region to be focused can be select by the line of sight. The image capturing apparatus **1000** of the present embodiment is provided with a line-of-sight operation unit

1016 with which a region selection operation by the line of sight can be performed to determine the focus detection region of the camera. The image capturing apparatus **1000** performs focus detection on the region thus determined.
(53) As described in the first to the fourth embodiments, the line-of-sight detection apparatus **901** used for the image capturing apparatus **1000** of the present embodiment can have a small size and perform line-of-sight detection with high accuracy that is less likely to be affected by the individual differences among people. Thus, the accuracy and speed of a series of operations performed by the image capturing apparatus **1000** of the present embodiment such as autofocusing and subject selection can be improved.
Other Embodiments
(54) Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.
(55) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(56) This application claims the benefit of Japanese Patent Application No. 2021-157238, filed Sep. 27, 2021 which is hereby incorporated by reference herein in its entirety.

Claims

1. A line-of-sight detection apparatus comprising: at least one processor or circuit configured to function as: an acquisition unit configured to acquire an eyeball image; an irradiation unit configured to irradiate an eyeball with light in an infrared wavelength band, forming a plurality of point-shaped optical images on the eyeball image; and a calculation unit configured to calculate line-of-sight information based on the eyeball image and the plurality of point-shaped optical images, wherein the irradiation unit includes at least one single light source configured to emit the light in the infrared wavelength band, and a dividing element configured to divide the light emitted from the single light source into a plurality of beams of the emitted light to form the plurality of point-shaped optical images, wherein the irradiation unit forms pairs of the point-shaped optical images, at least in a horizontal direction and a vertical direction, on the eyeball image, the point-shaped optical images in each of the pairs being opposite to each other with the center of the eyeball image interposed in between.
2. The line-of-sight detection apparatus according to claim 1, wherein the irradiation unit forms the

plurality of point-shaped optical images two-dimensionally arranged on the eyeball image.

3. The line-of-sight detection apparatus according to claim 1, wherein the irradiation unit forms the plurality of point-shaped optical images, in such a manner that even when part of the plurality of point-shaped optical images is hidden by an eyelid, remaining ones of the plurality of point-shaped optical images are two-dimensionally arranged.

4. The line-of-sight detection apparatus according to claim 1, wherein the irradiation unit forms the optical images on a straight line passing through the center of the eyeball image.

5. An image capturing apparatus comprising: an image capturing device configured to capture an image of a subject; a display apparatus configured to be able to display the image captured by the image capturing device; and the line-of-sight detection apparatus according to claim 1.

6. The image capturing apparatus according to claim 5, wherein the display apparatus is further able to display the line-of-sight information.

7. The image capturing apparatus according to claim 5, wherein an image on the display apparatus is observable by a user through the line-of-sight detection apparatus.